📝 Unit 1 Classwork — Real Numbers

Converting · Classification · Squares & Cubes · Estimating Roots · Ordering

A. Converting Fractions ↔ Decimals

Problem 1

Convert $\frac{3}{8}$ to a decimal.

0.375

Problem 2

Convert $\frac{5}{6}$ to a decimal (round to the thousandths, use a repeating bar notation if needed).

0.83^{-}

Problem 3

Convert 0.45 to a fraction in simplest form.

$\frac{9}{20}$

Problem 4

Convert $\frac{7}{20}$ to a decimal.

0.35

Problem 5

Convert $0.\overline{6}$ (repeating) to a fraction.

$\frac{2}{3}$

Problem 6

Convert $\frac{1}{3}$ to a decimal.

0.3^{-}

B. Classifying Numbers

Problem 1

Is √16 rational or irrational?

rational

Problem 2

Is $\sqrt{7}$ rational or irrational?

irrational

Problem 3

Is 0.454545... (repeating) rational or irrational?

rational

Problem 4

Is π rational or irrational?

irrational

Problem 5

Is -5 a whole number, an integer, or both?

integer

Problem 6

Is $\sqrt{36}$ rational or irrational?

rational

C. Perfect Squares & Cubes

Problem 1

What is 9^2?

81

Problem 2

What is 12^2?

144

Problem 3

What is 5^3?

125

Problem 4

What is $\sqrt{121}$?

11

Problem 5

What is $\sqrt[3]{64}$?

4

Problem 6

What is $\sqrt{225}$?

15

Problem 7

What is $\sqrt[3]{27}$?

3

D. Estimating Square & Cube Roots

Problem 1

Between which two consecutive integers does $\sqrt{50}$ fall?

7 and 8

Problem 2

Estimate $\sqrt{50}$ to the nearest tenth.

7.1

Problem 3

Between which two consecutive integers does $\sqrt[3]{100}$ fall?

4 and 5

Problem 4

Between which two consecutive integers does $\sqrt{90}$ fall?

9 and 10

E. Ordering Rational & Irrational Numbers

Problem 1

Order from least to greatest: $\sqrt{10}$, 3, $\frac{10}{3}$

$\sqrt{10}$, 3, $\frac{10}{3}$

Problem 2

Which is greater: $\sqrt{2}$ or $\frac{3}{2}$?

$\frac{3}{2}$

Problem 3

Which is greater: π or $\sqrt{10}$?

π